

OSI LAYERS

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

Layer	Name	Example Device/Protocol
7	Application	HTTP, FTP, SMTP
6	Presentation	SSL/TLS, JPEG
5	Session	NetBIOS, RPC
4	Transport	TCP, UDP
3	Network	IP, Router
2	Data Link	Switch, Ethernet
1	Physical	Hub, Network Cable

2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.
Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

- a) Encrypting data before sending: **Presentation Layer (Layer 6)**
- b) Routing packets across networks: **Network Layer (Layer 3)**
- c) Converting data to electrical signals: **Physical Layer (Layer 1)**
- d) Opening a web page in your browser: **Application Layer (Layer 7)**
- e) Detecting and correcting errors in frames: **Data Link Layer (Layer 2)**

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Example: WhatsApp

When I send a message on WhatsApp, different OSI layers work together:

Application Layer (Layer 7) – WhatsApp lets me type and send messages.

Presentation Layer (Layer 6) – The message is encrypted to keep it secure.

Transport Layer (Layer 4) – TCP ensures the message reaches the receiver correctly.

Network Layer (Layer 3) – IP routes the message through the internet.

Physical Layer (Layer 1) – Wi-Fi or mobile data sends the signals between devices.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Possible Problem Layers

Application Layer (Layer 7)

- Ping uses ICMP and only checks connectivity.
- The web server may not be running HTTP/HTTPS services.
- Browser-related issues may prevent loading the website.

**Presentation Layer (Layer 6)**

- SSL/TLS certificate problems can block HTTPS access.
- Encryption or certificate errors may stop the browser from opening the site.

Session Layer (Layer 5)

- Problems establishing or maintaining the communication session can prevent webpage access.

Reasoning

Since the IP address responds to ping:

- Physical Layer is working.
- Data Link Layer is working.
- Network Layer is working.
- Basic connectivity exists.